University of Utah Fall 2025

MATH 3220-002 PSet 12 Specification

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1 Background

This is the last problem set of MATH 3220. The focus is on the integration of differential 1-forms over 1-chains, as well as the consequences of the Stokes theorem for 1-chains (aka the Fundamental Theorem of Line Integrals).

Orientation Let V be a finite dimensional normed space and put $d = \dim(V)$. Suppose $d \geq 1$ and consider the product normed space $V^d = V \times V \times \cdots \times V$. Thus any d-tuple of vectors of V is an element of V^d; in particular any basis of V is an element of V^d.

The subset $\mathrm{Bases}(V) \subseteq V^d$ of all bases of V is an open subset of V^d, say as a consequence of problem 4 in problem set 5. Furthermore, $\mathrm{Bases}(V)$ has exactly two path-connected components. Any one of these two path-connected components is called an **orientation** on V. V is called **oriented** if one of the two orientations are specified.

Right Hand Rule versus Left Hand Rule If $V = \mathbb{R}^n$, adhering to the **right hand rule** (or the **counter clockwise direction**) in calculations means that the specified orientation is the path-connected component containing the standard basis; we denote this orientation by $\oplus$. Similarly the **left hand rule** (or the **clockwise direction**) is the other path-connected component; we denote this orientation by $\ominus$.

Let us define the **sign function** for orientations; this will make the integral formulas slightly more convenient to write later:

$$\begin{aligned} \operatorname{sign} :& \{\oplus, \ominus\} \to \{\pm 1\} \\ & ? \mapsto \begin{cases} +1 & , \text{ if } ? = \oplus \\ -1 & , \text{ if } ? = \ominus \end{cases} . \end{aligned}$$

Cells Let $\ell \in \mathbb{Z}_{\geq 0}$. Then a $C^{\ell+1}$ p-**cell** $C = [\varphi; B, ?]$, where

- $B \subseteq \mathbb{R}^p$ is a standard aligned box,
- $\varphi \in C^{\ell+1}(\widetilde{B}; M)$ for some open subset $\widetilde{B} \subseteq \mathbb{R}^p$ such that $B \subseteq \widetilde{B}$,
- $? \in \{\oplus, \ominus\}$,

is a p-current on C^k differential p-forms defined by the following formula: for any $\omega \in C^k\text{-Form}^p(M)$:

$$\begin{aligned} \langle C \,|\, \omega \rangle &= \operatorname{sign}(?) \int_B \star\varphi^*(\omega)(x_1, ..., x_p) d(x_1, ..., x_p) \\ &= \begin{cases} \int_B \star\varphi^*(\omega)(x_1, ..., x_p) d(x_1, ..., x_p), & \text{if } ? = \oplus \\ -\int_B \star\varphi^*(\omega)(x_1, ..., x_p) d(x_1, ..., x_p), & \text{if } ? = \ominus \end{cases} , \end{aligned}$$

where the integrals involved are unsigned Riemann integrals as we discussed before in multivariable integral calculus. A $C^{\ell+1}$ p-**chain** on M is a finite linear combination of $C^{\ell+1}$ p-cells on M, all considered as p-currents.

Boundary of a 1-Cell The 1-cell $[\gamma; \beta \leftarrow \alpha]$ is about integration from α to β. The topological boundary is $\{\alpha, \beta\}$. As the integration is from α to β, the orientation of $[\alpha]$ ought to be inward and the orientation of $[\beta]$ ought to be outward. Hence

$$\partial[\gamma; \beta \leftarrow \alpha] = [\gamma(\beta)] - [\gamma(\alpha)].$$

(Of course, ultimately the reason why I interpret inward as - is because I want to be able to use the boundary operator in the Fundamental Theorem of Calculus.)

Boundary of a 2-Cell Let $C = [\gamma; B, ?]$ be a 2-cell. Thus $B = [\lambda_1, \rho_1] \times [\lambda_2, \rho_2]$. and the topological boundary of B consists of the union of the following:

- Bottom edge $[\lambda_1, \rho_1] \times \{\lambda_2\}$
- Right edge $\{\rho_1\} \times [\lambda_2, \rho_2]$
- Top edge $[\lambda_1, \rho_1] \times \{\rho_2\}$
- Left edge $\{\lambda_1\} \times [\lambda_2, \rho_2]$.

If $? = +$, B is oriented in the counter clockwise, and hence these four edges ought to be oriented so as to keep B always on the left when traversing any one of them in the chosen orientation. Thus

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$$
\begin{aligned}
\partial[\gamma; \mathrm{B}, \oplus] = {} & [\gamma;\ \text{right}, \oplus] \\
& - [\gamma;\ \text{left}, \oplus] \\
& \quad - [\gamma;\ \text{top}, \oplus] \\
& \quad\quad + [\gamma;\ \text{bottom}, \oplus] \\
= {} & [\gamma; \{\rho_1\} \times [\lambda_2, \rho_2], \oplus] \\
& - [\gamma; \{\lambda_1\} \times [\lambda_2, \rho_2], \oplus] \\
& \quad - [\gamma; [\lambda_1, \rho_1] \times \{\rho_2\}, \oplus] \\
& \quad\quad + [\gamma; [\lambda_1, \rho_1] \times \{\lambda_2\}, \oplus] \\
= {} & [\gamma; (\rho_1, \rho_2) \leftarrow (\rho_1, \lambda_2)] \\
& - [\gamma; (\lambda_1, \rho_2) \leftarrow (\lambda_1, \lambda_2)] \\
& \quad - [\gamma; (\rho_1, \rho_2) \leftarrow (\lambda_1, \rho_2)] \\
& \quad\quad + [\gamma; (\rho_1, \lambda_2) \leftarrow (\lambda_1, \lambda_2)] \\
= {} & [\gamma; (\rho_1, \lambda_2) \leftarrow (\lambda_1, \lambda_2)] \\
& + [\gamma; (\rho_1, \rho_2) \leftarrow (\rho_1, \lambda_2)] \\
& \quad + [\gamma; (\lambda_1, \rho_2) \leftarrow (\rho_1, \rho_2)] \\
& \quad\quad + [\gamma; (\lambda_1, \lambda_2) \leftarrow (\lambda_1, \rho_2)].
\end{aligned}
$$

If on the other hand ? = -, then all the sides should be oriented the other way around, so as to have B always on the right when traversing along the topological boundary.

2 What to Submit

Submit your detailed solutions to each of the problems below. While the prompts may seem lengthy, the additional text is intended to guide you by providing context and helpful details. The hyperlinks too are meant to serve as optional additional context; chasing and studying them are by no means necessary to solve the problems. Indeed this problem set is designed so that, broadly speaking, the lectures are sufficient to complete it.

When tackling statements in MATH 3220, follow these four steps (in no particular order):

1. **Assess the truth of the statement:** Determine, either exactly or probabilistically, whether the statement is true.

2. **Build a case:** Come up with examples or begin drafting a proof to identify potential issues or areas where the argument may break down.

3. **Construct a proof:** Provide a formal argument to support your conclusion.

4. **Perturb the statement:** Experiment with logical variations of the statement. Can you prove a stronger/weaker result, or find alternative proofs?

In this problem set, and in later problem sets as well as exams, your work must address the third step (aka constructing a proof). While work regarding the remaining three items likely will help you write a formal proof, understand the material as well as identify connections between different concepts, you do not need to turn in generalizations or multiple proofs of the same statement, nor do you need to provide examples or counterexamples (unless you are specifically asked to). However, demonstrating these steps can contribute to partial credit when appropriate.

Statements in problems will typically be presented in neutral language. For example, a problem may simply state "P" (for P a well-defined statement) rather than "Show that P" or "Prove that P is false".

Your task is to determine the truth of the statement and provide a formal proof or refutation.

If a statement is false, it is highly recommended to propose a corrected version of the statement and prove this corrected version. This process, part of "perturbing the statement", will enhance your mathematical error detection and correction skills.

Unless explicitly stated otherwise, all problems require proofs. If you provide an example (say as proof to a given existential statement), you must also prove that the example satisfies the necessary conditions.

Some problems may be (variants of) theorems or exercises from the textbook; it's up to you to locate them (if need be)!

Make sure that each solution is properly enumerated and organized. Start the solution to each problem on a new page, and consider using headings or subheadings to structure your work clearly. This will not only aid in your thought process but also ensure that no part of your solution is overlooked during grading.

1. **There is nothing to prove in this problem.** Let $M = \mathbb{R}^2$ and consider the following three C^∞ 1-cells:

   - $\gamma_1 : t \mapsto (2t-1, 0)$, $C_1 = [\gamma_1; 1 \leftarrow 0]$
   - $\gamma_2 : t \mapsto (1-t, t)$, $C_2 = [\gamma_2; 1 \leftarrow 0]$
   - $\gamma_3 : t \mapsto (t-1, t)$, $C_3 = [\gamma_3; 1 \leftarrow 0]$

   Perform the following tasks:

   (a) Find all $n = (n_1, n_2, n_3) \in \mathbb{Z}^3$ with $|n_1| + |n_2| + |n_3| \leq 3$ such that the C^∞ 1-chain $n_1C_1 + n_2C_2 + n_3C_3$ is a boundary.

   (b) Find all $n = (n_1, n_2, n_3) \in \mathbb{Z}^3$ with $|n_1| + |n_2| + |n_3| \leq 3$ such that the C^∞ 1-chain $n_1C_1 + n_2C_2 + n_3C_3$ is a cycle.

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2. **There is nothing to prove in this problem.** Let $\gamma_1 : t \mapsto (2t-1, 0)$, $\gamma_2 : t \mapsto (\cos(\pi t), \sin(\pi t))$ and consider the C^∞ 1-chain $C = [\gamma_1; 1 \leftarrow 0] + [\gamma_2; 1 \leftarrow 0]$. Compute

$$\langle\, C \,|\, \omega + \eta \,\rangle,$$

where

$$\omega = 3x^2 dx + 2y dy, \qquad \eta = x dy.$$

3. Determine the number of distinct C^∞ 1-chains on $\mathbb{R}^2$ in the following list:

   - $[\gamma_1; 1 \leftarrow 0]$, where $\gamma_1 : t \mapsto (t^3, t^2)$
   - $[\gamma_2; \pi/2 \leftarrow 0]$, where $\gamma_2 : t \mapsto (\sin^3 t, 1 - \cos^2 t)$
   - $[\gamma_3; 2\pi \leftarrow 0]$, where $\gamma_3 : t \mapsto (\cos t, \sin t)$
   - $[\gamma_3; 4\pi \leftarrow 0]$
   - $[\gamma_3; 0 \leftarrow -\pi]$
   - $[\gamma_4; 1 \leftarrow 0]$, where $\gamma_4 : t \mapsto (t, \sqrt{1 - t^2})$

4. Let $M \subseteq E$ be an open subset, ω be an exact C^0 differential 1-form on M. Then for any $\gamma \in C^1(\mathbb{R}; M)$ and for any $\alpha, \beta \in \mathbb{R}$,

$$\gamma(\alpha) = \gamma(\beta) \implies \langle\, [\gamma; \beta \leftarrow \alpha] \,|\, \omega \,\rangle = 0.$$

5. Let $M \subseteq \mathbb{R}^2$ be an open subset not containing $(0, 0)$, and consider the following C^∞ differential 1-form on M:

$$\omega = \frac{-y}{x^2 + y^2} dx + \frac{x}{x^2 + y^2} dy.$$

(a) ω is closed.

(b) ω is exact.

3 Generative AI and Computer Algebra Systems Regulations

This section applies if you decide to use either a generative AI tool or a computer algebra system for this problem set. If not, you may skip this section.

3.1 Submission Requirements: Interaction Logs

If you use such tools, you must submit a complete log of your interactions. This is a mandatory part of your submission.

You can fulfill this requirement by either:

- **Listing the URLs** for your interactions in the appropriate field of the submission form, or
- **Attaching PDFs** containing the full interaction history to your Gradescope submission.

How to Obtain Logs You can generate the necessary logs using one of the following methods:

- **Shared Links:** Some AI tools (e.g., ChatGPT) can generate a unique, shareable URL for a conversation.
- **Web Archives:** Use a service like the Wayback Machine (see their "Save Page Now" documentation) to create a permanent, timestamped archive of your chat. There are also third-party

browser extensions that are designed to extract chat logs from generative tools.

- **Screenshots:** Manually take screenshots of the entire interaction and compile them into a single PDF file. You can use your operating system's built-in tools or a third-party application.

3.2 Chat Guidelines and Prompt Engineering

When using chatbots, you must adhere to the following guidelines:

- **Permitted:** You may copy and paste text from this problem set, the textbook, or other course materials into the chat to provide context.
- **Prohibited:** You may not directly ask for a complete solution to a problem.
- **Required:** You must begin your conversation with a "guardrails" prompt designed to structure the AI's responses. This practice is known as prompt engineering, and it is your responsibility you design an appropriate guardrails prompt. The staff will evaluate the reasonableness of your attempts based on the logs you submit.

  Here are some guides regarding prompt engineering:

  – https://platform.openai.com/docs/guides/prompt-engineering
  – https://developers.google.com/machine-learning/resources/prompt-eng
  – https://www.ibm.com/topics/prompt-engineering
  – https://aws.amazon.com/what-is/prompt-engineering/

When in doubt, check with the course staff to ensure your approach is appropriate.

4 How to Submit

- **Step 1 of 2:** Submit the form at the following URL:

  https://forms.gle/dheFCrdFcpwCJmZ98.

  You will receive a zero for this assignment if you skip this step, even if you submit your work on Gradescope on time.

- **Step 2 of 2:** Submit your work on Gradescope at the following URL:

  https://www.gradescope.com/courses/1073533/assignments/6451787,

  see the Gradescope documentation for instructions.

  You will receive a zero for this assignment if you submit your work on Gradescope under the wrong assignment, even if you submit your work on time.

5 When to Submit

This problem set is due on December 7, 2025 at 11:59 PM.

As per the course's syllabus, late submissions, up to 24 hours after this deadline, will be accepted with a 10% penalty. Submissions more than 24 hours late will not be accepted unless you contact the course staff with a valid excuse before the 24-hour extension expires.